

ZOMATO DATA ANALYSIS

BUSINESS INTELLIGENCE AND ANALYTICS

PYTHON | SQL | POWER BI

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EXECUTIVE OVERVIEW

Business Objective

Analyze operational, pricing, customer satisfaction, and delivery performance trends across 44,891 restaurants to identify key drivers influencing restaurant success and customer experience.

44.891K
Total Restaurants

3.92
Average Rating

33.00
Average Delivery Time

173.00
Average Price

Key Strategic Insights

- Moderate-rated restaurants represent nearly 53% of the platform, indicating significant service quality standardization opportunities.
- Fast delivery restaurants achieve the highest customer satisfaction with average ratings above 4.0.
- Kolkata, Kanpur, and NCR demonstrate the highest restaurant concentration, highlighting key competitive markets.
- Premium-priced restaurants do not consistently achieve higher customer ratings, indicating pricing inefficiencies.

Dataset

- 44,891 Restaurants
- Multi-city India dataset
- Key Columns include: Ratings, pricing, cuisines, delivery metrics

Tools Used

- Python (EDA & Feature Engineering)
- SQL (Business Querying)
- Power BI (Dashboarding & Visualization)

Methodology

- Data Cleaning
- Exploratory Data Analysis
- Feature Engineering
- SQL-Based KPI Analysis
- Interactive Dashboard Development

PROJECT WORKFLOW



Tools and Technologies

Python | Pandas | NumPy | SQL | Power BI | Matplotlib | Seaborn

DATASET OVERVIEW

Dataset Summary

- Source: Kaggle Zomato Dataset
- Scope: Multi-city Indian Restaurants
- Total Records: 44,891
- Dataset Type: Structured Tabular Data
- Tools Used: Python, SQL, Power BI
- Key Analysis Areas: Ratings, Pricing, Delivery, Customer Experience

Column Name	Description
Restaurant Name	Restaurant Name
Location	City / Area
Rating	Customer Rating
Average Price	Average Meal Cost
Average Delivery Time	Average Delivery Duration
Cuisine	Food Category
Safety Measure	Safety Compliance Status

Feature Engineering

- Rating Category : Highly Rated, Moderate Rated, Low Rated
- Price Category : Low, Medium, High
- Delivery Speed Category : Fast , Moderate, Slow
- City-Wise Operational Metrics
- Restaurant Performance Segmentation

Purpose

- Improve segmentation analysis
- Enable dashboard filtering
- Support operational analysis
- Improve business reporting

Missing Value
Handling

Duplicate
Removal

Data
Standardization

Data Type
Correction

PYTHON EDA & DATA CLEANING

Python Workflow

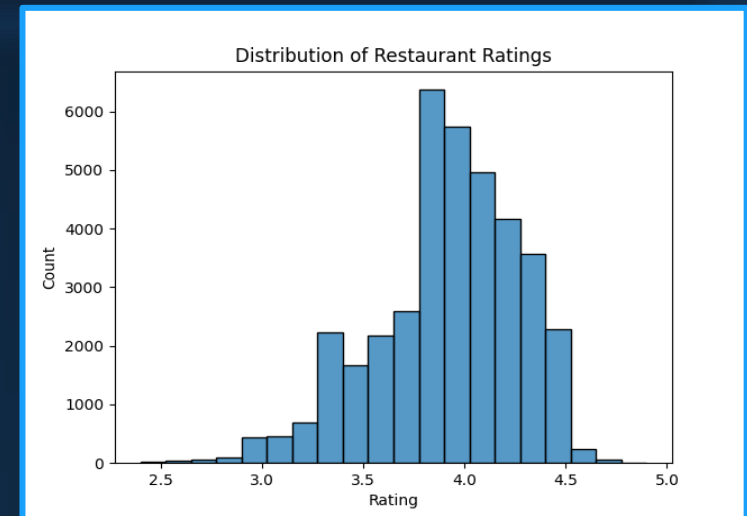
- Imported and explored raw Zomato dataset
- Performed missing value treatment and duplicate removal
- Standardized categorical and numerical variables
- Created engineered features for: Rating Category, Price Category, Delivery Speed Category
- Conducted EDA using Pandas, NumPy, Matplotlib, and Seaborn
- Exported cleaned datasets for SQL and Power BI analysis

Key EDA Insights

- Moderate-rated restaurants represented the largest segment of the dataset (~53%)
- Fast delivery restaurants achieved the highest average customer rating (4.03)
- Average delivery time across the platform was approximately 33 minutes
- Average meal pricing remained concentrated within low-to-medium price segments
- Significant operational variations were identified across cities and cuisines
- Feature engineering improved dashboard filtering, segmentation, and operational analysis

```
Final Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44891 entries, 0 to 44890
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Restaurant Name        44891 non-null  object
1   Rating                 37826 non-null  float64
2   Cuisine                44891 non-null  object
3   Average Price          44891 non-null  float64
4   Average Delivery Time  44891 non-null  float64
5   Safety Measure         44891 non-null  object
6   Location               44891 non-null  object
7   Is_Duplicate           44891 non-null  bool
dtypes: bool(1), float64(3), object(4)
memory usage: 2.4+ MB
None

Missing Values After Cleaning:
Restaurant Name    0
Rating             7065
Cuisine            0
Average Price      0
Average Delivery Time  0
Safety Measure     0
Location           0
Is_Duplicate       0
dtype: int64
```



Python • Pandas • NumPy • Matplotlib • Seaborn

SQL BUSINESS ANALYSIS

SQL Workflow

- Imported cleaned Zomato dataset into SQL database
- Performed KPI-driven business analysis using SQL queries
- Analyzed:
 - Restaurant performance
 - Delivery efficiency
 - Pricing trends
 - City-level operations
 - Customer ratings
- Applied:
 - GROUP BY
 - ORDER BY
 - CASE WHEN
 - Aggregate Functions
 - Filtering & Ranking
- Generated operational and strategic business insights for dashboard reporting

Business Questions Solved

- Which cities have the highest restaurant concentration?
- Which delivery category achieves the highest customer ratings?
- How does pricing influence customer satisfaction?
- Which cuisines experience longer delivery durations?
- Which operational segments require improvement?
- Which restaurant categories dominate the platform?
- How do delivery speed and ratings correlate?

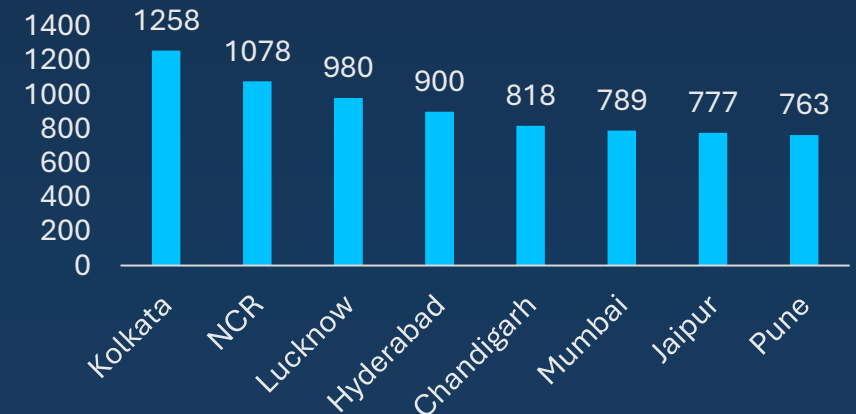
Key SQL Insights

- Fast delivery restaurants achieved the highest average ratings (~4.03)
- Moderate delivery segments represented nearly 72% of all restaurants
- Premium pricing did not consistently result in higher customer satisfaction

Top Performing Cities by Avg Rating



Restaurant Concentration by City



POWER BI DASHBOARD OVERVIEW

Dashboard Objectives

- Developed an interactive executive-level restaurant analytics dashboard
- Monitored customer satisfaction, delivery efficiency, and pricing behavior
- Enabled city-level operational benchmarking and performance comparison
- Created KPI-driven decision support for marketplace intelligence

Key Dashboard Features

- Interactive filters and slicers
- Dynamic customer experience analysis
- Restaurant pricing and rating intelligence
- Delivery performance tracking across cities
- Operational performance visualization

BI Tools & Technologies

- Power BI Desktop • DAX Measures & Calculated Columns
- Interactive Data Modeling • KPI Visualization Framework

ZOMATO DATA ANALYSIS



Key Dashboard Insight

The Power BI dashboard transformed cleaned restaurant operational data into an interactive executive intelligence platform for monitoring customer experience, delivery efficiency, pricing trends, and city-level performance.

EXECUTIVE KPI DASHBOARD

Executive Business Insights

Marketplace Scale

- Platform contains 44.8K+ restaurants across multiple Indian cities.

Customer Satisfaction

- Average customer rating remains strong at 3.92/5 indicating stable service quality.

Delivery Efficiency

- Average delivery time maintained at 33 minutes across the marketplace.

Pricing Benchmark

- Average order pricing is ₹173, positioning the platform within affordable mass-market segments.

Operational Observation

- Moderate-rated restaurants dominate the platform, representing the largest customer segment.

ZOMATO DATA ANALYSIS

44.891K

Total Restaurants

3.92

Average Rating

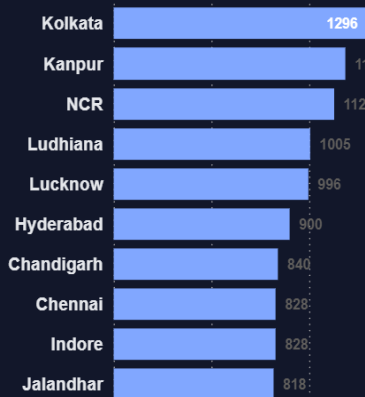
33.00

Average Delivery Time

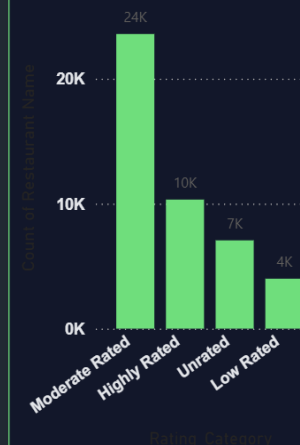
173.00

Average Price

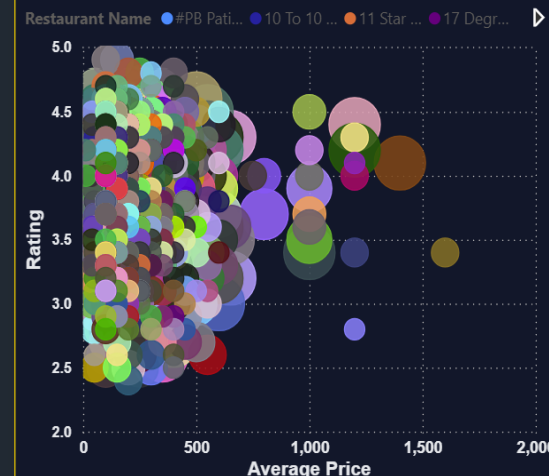
Top 10 Cities by Number of Restaurants



Distribution of Restaurants by Rating Category



Price vs Rating Relationship

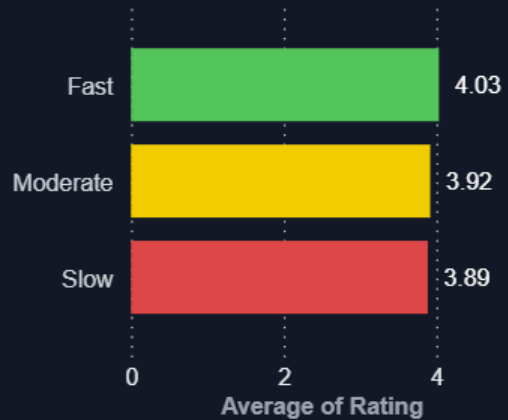


Key Executive Takeaway

The executive KPI dashboard provides a centralized operational view of restaurant performance, customer satisfaction, pricing behavior, and delivery efficiency, enabling data-driven strategic monitoring across the Zomato marketplace ecosystem.

CUSTOMER EXPERIENCE & RATING ANALYSIS

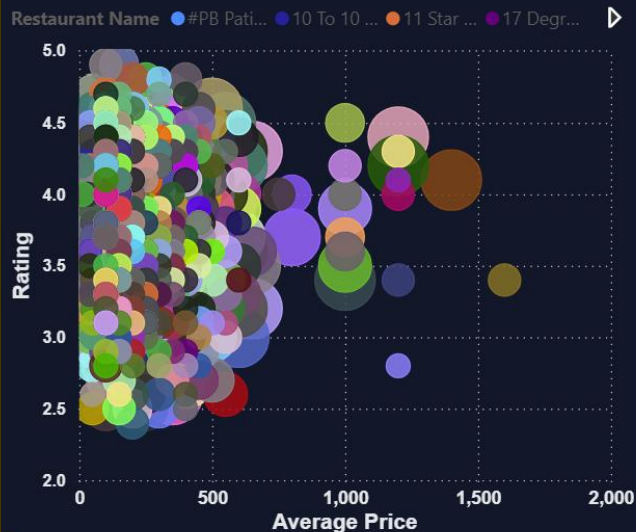
Average Customer Rating by Delivery Speed



Customer Experience Insights

- Fast delivery restaurants achieve the highest average customer rating (4.03).
- Slow delivery experiences reduce average customer ratings to 3.89.
- Moderate-rated restaurants dominate the platform and represent the largest operational segment.
- Customer ratings remain relatively stable within the 3.9–4.0 range across most cities.
- Higher pricing does not consistently guarantee higher customer satisfaction.
- Operational efficiency impacts customer experience more strongly than pricing alone.

Price vs Rating Relationship



Key Customer Experience Takeaway

Customer satisfaction across the Zomato ecosystem is driven more strongly by delivery efficiency and operational consistency than by premium pricing. Faster service quality directly contributes to improved restaurant ratings and customer experience outcomes.

Top Restaurant City

Kolkata • 1296

Highest Average Rating

Chennai • 4.02

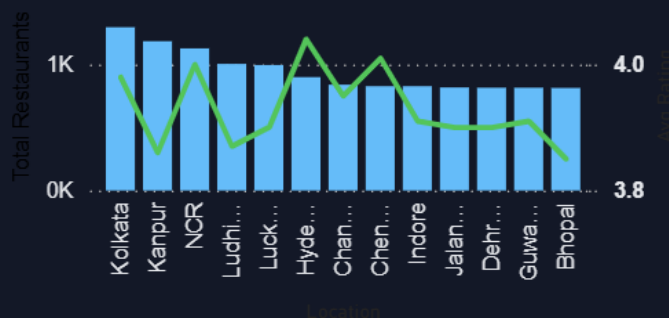
Fastest Delivery

31 Minutes Average

CITY-LEVEL OPERATIONAL PERFORMANCE

City-wise Restaurant Volume vs Average Rating

● Total Restaurants ● Avg Rating



Operational City Insights

- Kolkata leads the marketplace with the highest restaurant concentration (1296 restaurants).
- Chennai maintains the strongest average customer ratings among major operational cities.
- High-volume cities generally maintain stable customer ratings between 3.9–4.0.
- Operational scale does not necessarily reduce customer satisfaction performance.
- Delivery efficiency and operational consistency remain critical drivers of city-level ratings.
- Rapid-growth cities present expansion opportunities for delivery optimization initiatives.

Operational Risk & Opportunity Matrix by City

Location	Fast		Moderate		Slow	
	Average of Rating	Restaurant Count	Average of Rating	Restaurant Count	Average of Rating	Restaurant Count
Agra	4.06	18	3.90	408	3.75	174
Ahmedabad	4.13	58	4.00	448	3.89	94
Ajmer	3.88	62	3.87	340	3.87	14
Alappuzha	4.23	5	3.97	42	4.03	7
Aligarh	3.99	77	3.85	239	3.76	8
Allahabad	4.09	67	3.92	452	3.72	129
Amravati	3.93	70	3.86	261	3.76	21
Amritsar	3.98	94	3.79	476	3.76	54
Asansol	3.88	13	3.95	84	3.73	11
Aurangabad	3.88	5	3.90	102	3.70	5
Bangalore	4.03	21	4.12	223	4.00	290
Bareilly	4.11	72	3.84	443	3.65	40
Total	4.03	3600	3.92	32305	3.69	8976

Top Restaurant Hub

Kolkata • 1296

Best Rated City

Chennai • 4.02

Operational Rating Range

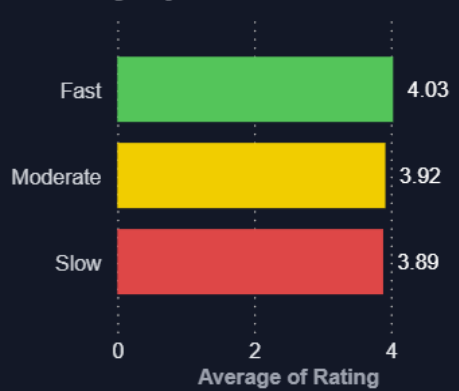
3.9 – 4.0

Key Operational Takeaway

City-level operational performance across the Zomato ecosystem demonstrates that large restaurant marketplaces can maintain stable customer satisfaction when supported by efficient delivery operations and scalable service management strategies.

DELIVERY PERFORMANCE & OPERATIONS ANALYSIS

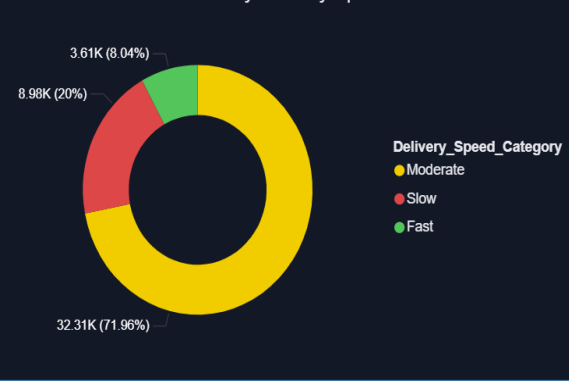
Average Customer Rating by Delivery Speed



Operational Delivery Insights

- Fast delivery restaurants achieve the strongest customer satisfaction performance with average ratings above 4.0.
- Slow delivery operations consistently reduce customer experience scores and operational reliability perception.
- Delivery efficiency demonstrates stronger influence on customer ratings than premium pricing alone.
- Moderate delivery segments dominate overall marketplace operations and represent the largest operational concentration.
- Stable delivery operations are critical for maintaining customer retention and service consistency across cities.
- Operational scalability requires balancing restaurant volume growth with delivery infrastructure efficiency.

Distribution of Restaurants by Delivery Speed



Operational Risk & Opportunity Matrix by City

Delivery_Speed_Category Location	Fast		Moderate		Slow	
	Average of Rating	Restaurant Count	Average of Rating	Restaurant Count	Average of Rating	Restaurant Count
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Bangalore	4.03	21	4.12	223	4.00	290
Bareilly	4.11	72	3.84	443	3.65	40
Total	4.03	3000	3.92	5200	3.89	1800

Fast Delivery Rating

4.03 Average

Moderate Delivery Share

~72% Restaurants

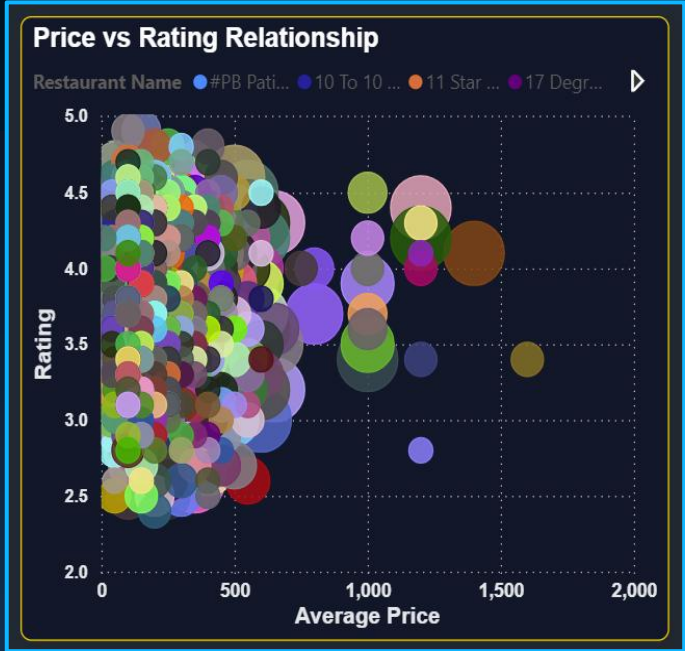
Slow Delivery Rating

3.89 Average

Key Operational Takeaway

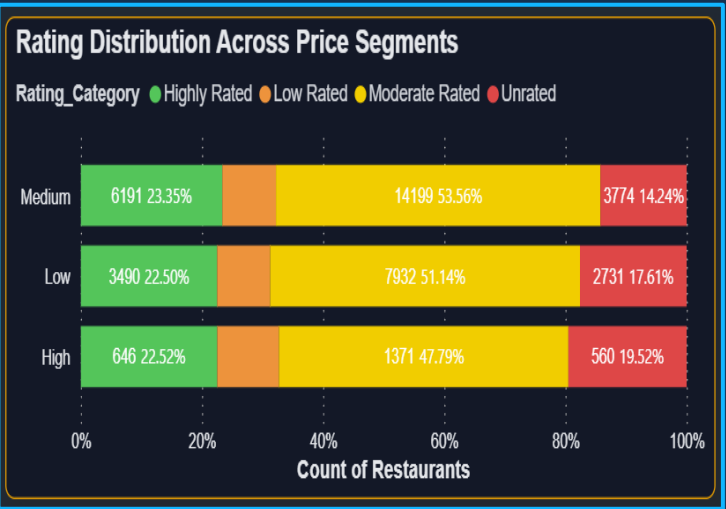
Operational efficiency and delivery consistency are the strongest marketplace performance drivers across the Zomato ecosystem. Faster delivery operations directly contribute to stronger customer satisfaction, while operational delays reduce service reliability and customer experience quality.

PRICING & VALUE INTELLIGENCE



Pricing Intelligence Insights

- Moderate-price restaurant segments dominate marketplace demand and operational concentration.
- Premium restaurant pricing does not consistently guarantee stronger customer satisfaction performance.
- Customer ratings remain relatively stable across multiple pricing segments, typically within the 3.9–4.0 range.
- Operational efficiency and service quality influence customer satisfaction more strongly than pricing alone.
- Affordable pricing strategies contribute significantly to marketplace scalability and customer retention.
- Pricing intelligence can help identify premium restaurants underperforming on customer experience metrics.



Average Marketplace Price

₹173 Average

Dominant Price Segment

Moderate Pricing

Customer Rating Stability

~3.9–4.0 Range

Key Pricing Takeaway

Customer satisfaction across the Zomato ecosystem is influenced more strongly by operational consistency and delivery efficiency than premium pricing alone. Moderate and affordable pricing segments continue to drive marketplace scale, customer engagement, and operational sustainability.

STRATEGIC BUSINESS RECOMMENDATIONS

Delivery Optimization Strategy

- Introduce city-level delivery optimization initiatives in high-volume operational hubs to improve delivery efficiency and customer retention.
- Prioritize operational monitoring for cities with growing restaurant concentration and slower delivery segments.

Customer Experience Enhancement

- Standardize service quality benchmarking across restaurant partners to reduce rating volatility.
- Encourage low-rated restaurant segments to adopt faster fulfillment and quality-control mechanisms.

Operational Expansion Strategy

- High-volume cities like Kolkata demonstrate strong scalability potential for marketplace expansion.
- Identify rapidly growing cities for operational investment and delivery infrastructure scaling.

Pricing & Value Intelligence

- Expand value-driven pricing strategies to strengthen customer affordability perception.
- Use pricing intelligence to identify premium restaurants underperforming on customer experience metrics.

Strategic Business Conclusion

Customer experience, delivery efficiency, and operational consistency emerged as the strongest marketplace performance drivers across the Zomato ecosystem. Strategic investment in delivery optimization, operational benchmarking, and value-focused customer experience initiatives can significantly improve platform scalability, customer retention, and long-term marketplace performance.

PROJECT CONCLUSION & BUSINESS IMPACT

Project Summary

- Successfully completed an end-to-end restaurant marketplace analytics project using Python, SQL, and Power BI.
- Analyzed 44,891 restaurant records across multiple Indian cities to identify customer experience, delivery, pricing, and operational trends.
- Built a structured analytics workflow including data cleaning, EDA, feature engineering, SQL business intelligence, and interactive dashboard reporting.

Final Business Conclusion

The project demonstrates how data analytics can transform large-scale operational restaurant data into measurable business intelligence for customer experience optimization, operational scalability, and strategic marketplace growth.

Business Impact & Key Outcomes

- Identified delivery efficiency as the strongest operational driver influencing customer ratings and satisfaction.
- Observed that fast-delivery restaurants consistently achieved higher average ratings (~4.03) compared to slower delivery segments (~3.89).
- Identified Kolkata as the largest operational restaurant hub and Chennai as one of the strongest customer satisfaction performers.
- Developed executive-level Power BI dashboards enabling city-level benchmarking, KPI monitoring, and operational performance analysis.
- Generated strategic recommendations for delivery optimization, operational scaling, pricing intelligence, and customer experience enhancement.

Core Technologies

Python
SQL
Power BI
Pandas
NumPy
Matplotlib
Seaborn
DAX

Business Domains

Customer Analytics
Operational Intelligence
Delivery Performance
Pricing Analytics
Marketplace Strategy
Business Intelligence

KEY LEARNINGS & ANALYTICS SKILLS

Technical Skills Developed

- Python data cleaning and preprocessing workflows
- Exploratory data analysis and feature engineering
- SQL business intelligence querying and KPI analysis
- Power BI dashboard design and DAX implementation
- Data storytelling and executive KPI visualization
- End-to-end analytics workflow development

Key Project Learnings

- Operational efficiency strongly impacts customer satisfaction
- Faster delivery performance improves platform ratings
- Moderate pricing segments dominate marketplace demand
- Scalable analytics workflows improve decision-making
- Dashboard visualization enhances executive intelligence reporting

Business Intelligence Capabilities

- Customer experience and rating analysis
- Delivery performance monitoring and benchmarking
- Pricing intelligence and marketplace analysis
- City-level operational performance evaluation
- Strategic recommendation development
- KPI-driven business decision support

End-to-End Project Capabilities

- Built an end-to-end analytics project using marketplace data
- Strengthened practical problem-solving and business storytelling skills
- Developed dashboard presentation and executive reporting capabilities
- Applied KPI-driven analytical thinking for operational decision-making
- Gained hands-on experience across Python, SQL, and Power BI ecosystems